

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250287699

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

LIAW; Jhon Jhy

DUAL-PORT SRAM STRUCTURE

Abstract

The static random access memory (SRAM) cell of the present disclosure includes a first pull-down device, a second pull-down device, a first pass-gate device, and a second pass-gate device in a first p-well on a substrate; a third pull-down device, a fourth pull-down device, a third pass-gate device, and a fourth pass-gate device in a second p-well on the substrate; a first pull-up device and a second pull-up device in an n-well between the first p-well and the second p-well; and a first landing pad between the second pull-down device and the first pull-up device. The first landing pad is electrically coupled to a gate structure of the second pass-gate device by way of a first gate via.

Inventors: LIAW; Jhon Jhy (Hsinchu County, TW)

Applicant: Taiwan Semiconductor Manufacturing Company Limited (Hsin-Chu, TW)

Family ID: 76709282

Appl. No.: 19/218425

Filed: May 26, 2025

Related U.S. Application Data

parent US continuation 18439486 20240212 parent-grant-document US 12317601 child US 19218425

parent US continuation 17811260 20220707 parent-grant-document US 11901352 child US 18439486

parent US continuation 16932394 20200717 parent-grant-document US 11444072 child US 17811260

us-provisional-application US 62981317 20200225

Publication Classification

Int. Cl.: H10D89/10 (20250101); H10B10/00 (20230101); H10D84/85 (20250101)

Background/Summary

PRIORITY DATA [0001] This application is a continuation of U.S. patent application Ser. No. 18/439,486, filed Feb. 12, 2024, which is a continuation of U.S. patent application Ser. No. 17/811,260, filed Jul. 7, 2022, which is a continuation of U.S. patent application Ser. No. 16/932,394, filed Jul. 17, 2020, which claims priority to U.S. Provisional Patent Application No. 62/981,317 filed on Feb. 25, 2020, entitled “DUAL-PORT SRAM STRUCTURE”. U.S. patent application Ser. No. 18/439,486, U.S. patent application Ser. No. 17/811,260, U.S. patent application Ser. No. 16/932,394, and U.S. Provisional Patent Application No. 62/981,317 are incorporated herein by reference in their entireties.

BACKGROUND

[0002] The semiconductor industry has experienced rapid growth. Technological advances in semiconductor materials and design have produced generations of semiconductor devices where each generation has smaller and more complex circuits than the previous generation. In the course of integrated circuit (IC) evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. However, these advances have also increased the complexity of processing and manufacturing semiconductor devices.

[0003] In deep sub-micron integrated circuit technology, an embedded static random access memory (SRAM) device has become a popular storage unit of high speed communication, image processing and system-on-chip (SOC) products. For example, a dual port (DP) SRAM device allows parallel operation, such as 1R (read) 1 W (write), and 2R (read) or 2 W (Write) in one cycle, and therefore has higher bandwidth than a single port SRAM. In advanced technologies with decreased feature size and increased packing density, low loading and high speed of the cell structure are important factors in embedded memory and SOC products. Dual port SRAM cell may have complex metal routing that may pose challenges in routing arrangement and lithography process windows. Therefore, although existing DP SRAM cells are generally adequate for their intended purposes, they are not satisfactory in all aspects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 is a schematic circuit diagram of a dual port static random access memory (DP SRAM) cell constructed according to various aspects of the present disclosure.

[0006] FIG. 2 is an alternative schematic circuit diagram of a DP SRAM cell constructed according to various aspects of the present disclosure.

[0007] FIG. 3 is a schematic illustration of a DP SRAM array constructed according to various aspects of the present disclosure.

[0008] FIG. 4A is a schematic layout of a DP SRAM cell, according to various aspects of the present disclosure.

[0009] FIG. 4B is a simplified schematic layout of the DP SRAM cell in FIG. 4A, according to various aspects of the present disclosure.

[0010] FIG. 5 is a schematic illustration of various metal layers of a DP SRAM cell viewed along a direction of active regions, according to various aspects of the present disclosure.

[0011] FIG. 6 is a schematic illustration of various metal layers of a DP SRAM cell viewed along a direction of gate structures, according to various aspects of the present disclosure in one embodiment.

[0012] FIGS. 7, 8 and 9 illustrate embodiments of word lines of a DP SRAM cell, according to various aspects of the present disclosure.

DETAILED DESCRIPTION

[0013] The following disclosure provides many different embodiments, or examples, for implementing different features of the disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0014] In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Moreover, the formation of a feature on, connected to, and/or coupled to another feature in the present disclosure that follows may include embodiments in which the features are formed in direct contact, and may also include embodiments in which additional features may be formed interposing the features, such that the features may not be in direct contact. In addition, spatially relative terms, for example, “lower,” “upper,” “horizontal,” “vertical,” “above,” “over,” “below,” “beneath,” “up,” “down,” “top,” “bottom,” etc. as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) are used for ease of the present disclosure of one features relationship to another feature. The spatially relative terms are intended to cover different orientations of the device including the features.

[0015] Still further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range including the number described, such as within $\pm 10\%$ of the number described or other values as understood by person skilled in the art. For example, the term “about 5 nm” encompasses the dimension range from 4.5 nm to 5.5 nm.

[0016] FIGS. 1 and 2 are schematic circuit diagrams of a dual port static random access memory (DP SRAM) cell **100**. The DP SRAM cell **100** may be implemented using planar field effect transistors (FETs) or multi-gate FETs. A planar FET includes a gate structure that may induce a planar channel region along one surface of its active region, hence its name. A multi-gate FET includes a gate structure that is in contact with at least two surfaces of its active region. Examples of multi-gate FETs include fin-type FETs (FinFETs) and multi-bridge channel (MBC) FETs. A FinFET includes a fin-shaped active region arising from a substrate and a gate structure disposed over a top surface and sidewalls of the fin-shaped active region. A MBC FET includes at least one channel member extending between two source/drain features and a gate structure that wraps completely around the at least one channel member. Because of the feature that its gate structure

wraps around the channel member, an MBC FET may also be referred to as a gate-all-around (GAA) FET or a surrounding gate transistor (SGT). Depending on the shapes and orientation, a channel member in a MBC FET may be referred to as a nanosheet, a semiconductor wire, a nanowire, a nanostructure, a nano-post, a nano-beam, or a nano-bridge. In some instances, an MBC FET may be referred to by the shape of the channel member. For example, an MBC FET having one or more nanosheet channel member may also be referred to as a nanosheet transistor or a nanosheet FET. For avoidance of doubts, embodiments of the present disclosure may be applicable to DP SRAM cell **100** that is implemented using planar FETs or multi-gate FETs.

[0017] In some embodiments, the DP SRAM cell **100** may include a total of 10 transistors and may be referred to as a 10T SRAM cell **100**. Referring to FIG. **1**, the DP SRAM cell **100** includes a first inverter **101** and a second inverter **102** that are cross-coupled. As shown in FIG. **2**, the first inverter **101** includes a first pull-up device (PU-1) **142** formed with a p-type field-effect transistor (pFET), a first pull-down device (PD-1) **146** formed with an n-type FET (nFET), and a second pull-down device (PD-2) **148** formed with an nFET. The first pull-down device (PD-1) **146** and the second pull-down device (PD-2) **148** are configured in parallel mode. Specifically, the drains of the first pull-down device (PD-1) **146** and the second pull-down device (PD-2) **148** are electrically connected together, the corresponding sources thereof are electrically connected together, and the corresponding gates thereof are electrically connected together. The second inverter **102** includes a second pull-up device (PU-2) **144** formed with a pFET, a third pull-down device (PD-3) **150** formed with an nFET, and a fourth pull-down device (PD-4) **152** formed with an nFET. The third pull-down device (PD-3) **150** and the fourth pull-down device (PD-4) **152** are configured in parallel mode.

[0018] The drains of PU-1 (**142**), PD-1 (**146**) and PD-2 (**148**) are electrically connected together, defining a first drain node (also referred to as a first node or node **1**). The drains of PU-2 (**144**), PD-3 (**150**) and PD-4 (**152**) are electrically connected together, defining a second drain node (also referred to as a second node or node **2**). The gates of PU-1 (**142**), PD-1 (**146**) and PD-2 (**148**) are electrically connected and coupled to the second node. The gates of PU-2 (**144**), PD-3 (**150**) and PD-4 (**152**) are electrically connected and coupled to the first node. The sources of PU-1 (**142**) and PU-2 (**144**) are electrically connected to the power line (V_{dd} or CV_{dd}). The sources of PD-1 (**146**), PD-2 (**148**), PD-3 (**150**), and PD-4 (**152**) are electrically connected to a complementary power line (V_{ss} or CV_{ss}).

[0019] The DP SRAM cell **100** further includes a first port (port-A) **112** and a second port (port-B) **114**. In one embodiment, the port-A **112** and port-B **114** include at least four pass-gate devices, referred to as PG-1 (**104**), PG-2 (**106**), PG-3 (**108**), and PG-4 (**110**), respectively. The port-A **112** includes a first pass-gate device (PG-1) **104** and a second pass-gate device (PG-2) **106**. The port-B **114** includes a third pass-gate device (PG-3) **108** and a fourth pass-gate device (PG-4) **110**. The drain of PG-1 (**104**) is electrically connected to a first bit-line **122** (referred to as BL-A). The source of PG-1 (**104**) is electrically connected to the first node. The gate of PG-1 (**104**) is electrically connected to a first word-line **112** (referred to as port-A WL). The drain of PG-2 (**106**) is electrically connected to a first bit-line bar **124** (BL-A bar). The source of PG-2 (**106**) is electrically connected to the second node. The gate of PG-2 (**106**) is electrically connected to a first word-line **112** (port-A WL). The drain of PG-3 (**108**) is electrically connected to a second bit-line **132** (BL-B). The source of PG-3 (**108**) is electrically connected to the first node. The gate of PG-3 (**108**) is electrically connected to the second word-line **114** (port-B WL). The drain of PG-4 (**110**) is electrically connected to a second bit-line bar **134** (BL-B bar). The source of PG-4 (**110**) is electrically connected to the second node. The gate of PG-4 (**110**) is electrically connected to the second word-line **114** (port-B WL). It is noted that, for ease of reference, both port-A and port-A WL share the same reference number **112** and both port-B and port-B WL share the same reference numeral **114**. It can be seen from FIG. **2** that the DP SRAM cell **100** includes two pull-up devices (PU-1 and PU-2), four pull-down devices (PD-1, PD-2, PD-3, and PD-4), and four pass-gate

devices (PG-1, PG-2, PG-3, and PG-4), which may be implemented using 10 transistors (10T), such as 10 planar FETs or 10 multi-gate FETs.

[0020] FIG. 3 is a schematic illustration of a DP SRAM device 10. As shown in FIG. 3, the DP SRAM device 10 includes an SRAM array 12 comprised of 4×4 or more dual-port SRAM cells each of which may have the circuit layout shown in FIGS. 1 and 2. The first word-lines (WL-A) of the dual-port SRAM cells in the SRAM array 12 are connected to and controlled by a first word-line (WL-A) driver/selector 14, and the second word-lines (WL-B) of the dual-port SRAM cells in the SRAM array 12 are connected to and controlled by a second word-line (WL-B) driver/selector 16. According to various aforementioned embodiments, the first and second word-lines WL-A and WL-B of the dual-port SRAM cells in the SRAM array 12 are formed in the same or different metal layers, such as the second metal layer (M2) or and the fourth metal layer (M4). For example, the first word-line WL-A is formed in the second metal layer (M2) and the second word-line WL-B is formed in the fourth metal layer (M4). For another example, both the first word-line WL-A and the second word-line WL-B are formed in the second metal layer (M2). The different metal layers are illustrated in FIGS. 5 and 6 and will be described in more details below.

[0021] Still referring to FIG. 3, the first bit-line (BL-A) and the first bit-line bar (BL-A bar) of the dual-port SRAM cells in the SRAM array 12 are connected to a first Y-multiplexer (MUX) and sensing amplifier 18. When the first word-line WL-A is selected, the first bit-line (BL-A) and the first bit-line bar (BL-A bar) of an SRAM cell can be selected and data transmitted thereof can be read and sensed (or written) by the first Y-multiplexer (MUX) and sensing amplifier 18. The second bit-line (BL-B) and the second bit-line bar (BL-B bar) of the dual-port SRAM cells in the SRAM array 12 are connected to a second Y-multiplexer (MUX) and sensing amplifier 20. When the second word-line WL-B is selected, the second bit-line (BL-B) and the second bit-line bar (BL-B bar) of an SRAM cell can be selected and data transmitted thereof can be read and sensed (or written) by the second Y-multiplexer (MUX) and sensing amplifier 20. According to various embodiments according to the present disclosure, the first bit-line (BL-A), the first bit-line bar (BL-A bar), the second bit-line (BL-B), and the second bit-line bar (BL-B bar) of the dual-port SRAM cells can be formed in the same metal layer, such as the first metal layer (M1), shown in FIGS. 5 and 6.

[0022] To illustrate routing features in the DP SRAM cell 100 shown in FIGS. 1 and 2, schematic layout of a DP SRAM cell is illustrated in FIGS. 4A and 4B. Because the schematic layout in FIGS. 4A and 4B illustrates further features than the circuit diagrams in FIGS. 1 and 2, a different set of reference numerals is adopted for FIGS. 4A and 4B. It is noted, however, although different references numerals may be used in features in FIGS. 1, 2, 4A and 4B, devices and nodes identified in FIGS. 1 and 2 have corresponding counterparts in FIGS. 4A and 4B. For example, a DP SRAM cell, PG-1, PG-2, PG-3, PG-4, PD-1, PD-2, PD-3, PD-4, PU-1, PU-2, WL-A, WL-B, BL-A, BL-B, BL-A bar, BL-B bar, Vss, Vdd in FIGS. 1 and 2 correspond to similar features in FIG. 4A or 4B, despite of their different reference numerals. As compared to the schematic layout in FIG. 4A, the one shown in FIG. 4B is simplified to illustrate example inventive features of the present disclosure.

[0023] Reference is first made to FIG. 4A. A DP SRAM cell 100 shown in FIGS. 1 and 2 may be implemented as the DP SRAM cell 200 shown in FIG. 4A. The DP SRAM cell 200 may be fabricated on a substrate 202. In some embodiments, the substrate 202 may include a semiconductor material, such as silicon or silicon germanium. In some alternative embodiments, the substrate 202 may be a silicon-on-insulator (SOI) that include an insulating layer on a supporting bulk wafer for isolation and a silicon layer over the insulating layer. An SOI substrate may be formed using different techniques including Separation by Implanted Oxygen (SIMOX), Bonding and Etching Back SOI (BESOI), and Zone Melting and Recrystallization (ZMR) SOI. The substrate 202 may include different doped regions. In embodiments illustrated in FIG. 4A, with respect to the DP SRAM cell 200, the substrate 202 may include an n-type well 1000, which is

doped with an n-type dopant, such as phosphorus (P) and arsenide (As), and a first p-type well **2000-1** and a second p-type well **2000-2**, which are doped with a p-type dopant, such as boron (B). As will be described below, n-type devices are to be fabricated over the first and second p-type wells **2000-1** and **200-2**, and p-type devices are to be fabricated over the n-type well **1000**. In some implementations, when regarded beyond the boundary of the DP SRAM cell **200**, the first p-type well **2000-1** and the second p-type well **2000-2** may be different regions of the same p-type well **2000**. With respect to the DP SRAM cell **200**, the n-type well **1000** is sandwiched between the first p-type well **2000-1** and the second p-type well **2000-2**.

[0024] The DP SRAM cell **200** includes a first active region **204-1**, a second active region **204-2**, a third active region **204-3**, a fourth active region **204-4**, a fifth active region **204-5**, and a sixth active region **204-6**. Each of the active regions extend lengthwise along the Y direction. In some embodiments illustrated in FIG. 4A, each the first active region **204-1**, the second active region **204-2**, the fifth active region **204-5**, and the sixth active region **204-6** includes two fin structures or a two vertical stacks of channel members while each of the third active region **204-3** and the fourth active region **204-4** includes a single fin structure or a single vertical stack of channel members. Other numbers of active regions, arrangements and implementations are possible and are fully envisioned by the present disclosure. In some implementations, the first pull-up device (PU-1) **206** is formed over the third active region **204-3**, the second pull-up device (PU-2) **208** is formed over the fourth active region **204-4**, the first pass-gate device (PG-1) **210** and the first pull-down device (PD-1) **218** are formed over the first active region **204-1**, the second pass-gate device (PG-2) **212** and the second pull-down device (PD-2) **220** are formed over the second active region **204-2**, the third pull-down device (PD-3) **222** and the third pass-gate device (PG-3) **214** are formed over the fifth active region **204-5**, and the fourth pull-down device (PD-4) **224** and the fourth pass-gate device (PG-4) **216** are formed over the sixth active region **204-6**. In some instances illustrated in FIG. 4A, the first pass-gate device (PG-1) **210**, the second pass-gate device (PG-2) **212**, the third pass-gate device (PG-3) **214**, and the fourth pass-gate device (PG-4) **216** each have its own gate structure that extend lengthwise along the X direction, which is perpendicular to the Y direction. The first pull-down device (PD-1) **218**, the second pull-down device (PD-2) **220**, the first pull-up device (PU-1) **206** share one gate structure that also extends lengthwise along the X direction. The third pull-down device (PD-3) **222**, the fourth pull-down device (PD-4) **224**, the second pull-up device (PU-2) **208** share one gate structure that also extends lengthwise along the X direction.

[0025] The DP SRAM cell **200** includes gate vias and source/drain contacts form connections to the gate structures and sources/drains of the transistors therein. Referring to FIG. 4A, in some embodiments, the DP SRAM cell **200** includes a first gate via **252**, a second gate via **254**, a third gate via **256**, and a fourth gate via **258**. The first gate via **252** is disposed over and electrically coupled to the gate structure of the second pass-gate device (PG-2) **212**. Along the X direction, the first gate via **252** is disposed over an isolation feature between the second active region **204-2** and the third active region **204-3**. The isolation feature may be a shallow trench isolation (STI) feature and is representatively shown in FIG. 5. The first gate via **252** electrically couples the gate structure of the second pass-gate device (PG-2) **212** to a first WL-B landing pad **228-1**. The first WL-B landing pad **228-1** extends lengthwise along the Y direction and is situated between the second active region **204-2** and the third active region **204-3**. The second gate via **254** is disposed over and electrically coupled to the gate structure of the third pass-gate device (PG-3) **214**. Along the X direction, the second gate via **254** is disposed over an isolation feature (e.g., an STI feature) between the fourth active region **204-4** and the fifth active region **204-5**. The second gate via **254** electrically couples the gate structure of the third pass-gate device (PG-3) **214** to a second WL-B landing pad **228-2**. The second WL-B landing pad **228-2** extends lengthwise along the Y direction and is situated between the fourth active region **204-4** and the fifth active region **204-5**. The third gate via **256** is disposed over and electrically coupled to the gate structure of the first pass-gate device (PG-1) **210**. Along the X direction, the third gate via **256** is disposed at an end or boundary

of the DP SRAM cell **200**. The third gate via **256** electrically couples the gate structure of the first pass-gate device (PG-1) **210** to a first WL-A landing pad **226-1**. The first WL-A landing pad **226-1** extends lengthwise along the Y direction and is situated at an end or a boundary of the DP SRAM cell **200**. The fourth gate via **258** is disposed over and electrically coupled to the gate structure of the fourth pass-gate device (PG-4) **216**. Along the X direction, the fourth gate via **258** is disposed at the opposite end or boundary of the DP SRAM cell **200**. The fourth gate via **258** electrically couples the gate structure of the fourth pass-gate device (PG-4) **216** to a second WL-A landing pad **226-2**. The second WL-A landing pad **226-2** extends lengthwise along the Y direction and is situated at the opposite end or boundary of the DP SRAM cell **200**. Along the Z direction, each of the gate vias may have a circular shape, an oval shape, a racetrack shape, a square shape, or a rectangular shape. In some embodiments, because the third gate via **256** and the fourth gate via **258** are disposed at boundaries of the DP SRAM cell **200**, they may be enlarged to reduce resistance and improve performance of the DP SRAM cell **200**. On the contrary, the first gate via **252** and the second gate via **254** are disposed within active regions of the DP SRAM cell **200**, the room to enlarge them may be limited. As a result, in those embodiments, the X-Y plane projection area of the third gate via **256** and the fourth gate via **258** is greater than the X-Y plane projection area of the first gate via **252** and the second gate via **254**.

[0026] Drains (or drain features) of the first pull-down device (PD-1) **218**, the second pull-down device (PD-2) **220**, and the first pull-up device (PU-1) **206** are electrically coupled to a first long contact **240**. The first long contact **240** extends lengthwise along the X direction over the first active region **204-1**, the second active region **204-2**, and the third active region **204-3**. As shown in FIG. 4A, the first long contact **240** is disposed between the gate structure of the first pass-gate device (PG-1) **210** and the common gate structure of the first pull-down device (PD-1) **218**, the second pull-down device (PD-2) **220**, and the first pull-up device (PU-1) **206**. Similarly, the first long contact **240** is disposed between the gate structure of the second pass-gate device (PG-2) **212** and the common gate structure of the first pull-down device (PD-1) **218**, the second pull-down device (PD-2) **220**, and the first pull-up device (PU-1) **206**. The first long contact **240** is disposed between the common gate structure of the first pull-down device (PD-1) **218**, the second pull-down device (PD-2) **220**, and the first pull-up device (PU-1) **206** and the common gate structure of the third pull-down device (PD-3) **222**, the fourth pull-down device (PD-4) **224**, and the second pull-up device (PU-2) **208**. As shown in FIG. 4A, because the first long contact **240** extends along the X direction to span over the first WL-B landing pad **228-1** (which extends lengthwise along the Y direction), the first gate via **252** may be adjacent to the first long contact **240**. As used herein, a long contact refers to a contact feature that physically spans over more than one source/drain region. For example, the first long contact **240** physically spans over the drains of the first pull-down device (PD-1) **218**, the second pull-down device (PD-2) **220**, and the first pull-up device (PU-1) **206**. Alternatively, a long contact refers to a contact feature that physically spans over more than one active region. For example, the first long contact **240** physically spans over the first active region **204-1**, the second active region **204-2**, and the third active region **204-3**.

[0027] Drains (or drain features) of the third pull-down device (PD-3) **222**, the fourth pull-down device (PD-4) **224**, and the second pull-up device (PU-2) **208** are electrically coupled to a second long contact **242**. The second long contact **242** extends lengthwise along the X direction over the fourth active region **204-4**, the fifth active region **204-5**, and the sixth active region **204-6**. As shown in FIG. 4A, the second long contact **242** is disposed between the gate structure of the fourth pass-gate device (PG-4) **216** and the common gate structure of the third pull-down device (PD-3) **222**, the fourth pull-down device (PD-4) **224**, and the second pull-up device (PU-2) **208**. Similarly, the second long contact **242** is disposed between the gate structure of the third pass-gate device (PG-3) **214** and the common gate structure of the third pull-down device (PD-3) **222**, the fourth pull-down device (PD-4) **224**, and the second pull-up device (PU-2) **208**. The second long contact **242** is disposed between the common gate structure of the first pull-down device (PD-1) **218**, the

second pull-down device (PD-2) **220**, and the first pull-up device (PU-1) **206** and the common gate structure of the third pull-down device (PD-3) **222**, the fourth pull-down device (PD-4) **224**, and the second pull-up device (PU-2) **208**. As shown in FIG. 4A, because the second long contact **242** extends along the X direction to span over the second WL-B landing pad **228-2** (which extends lengthwise along the Y direction), the second gate via **254** may be adjacent to the second long contact **242**.

[0028] Sources (or source features) of the first pull-down device (PD-1) **218** and the second pull-down device (PD-2) **220** are electrically coupled to a third long contact **244**. The third long contact **244** extends lengthwise along the X direction over the first active region **204-1** and the second active region **204-2**. Sources (or source features) of the third pull-down device (PD-3) **222** and the fourth pull-down device (PD-4) **224** are electrically coupled to a fourth long contact **246**.

[0029] The fourth long contact **246** extends lengthwise along the X direction over the fifth active region **204-5** and the sixth active region **204-6**.

[0030] In addition to the first, second, third, and fourth long contacts, the first pass-gate device (PG-1) **210** has a first source contact **272**, the second pass-gate device (PG-2) **212** has a second source/drain contact **274**, the first pull-up device (PU-1) **206** has a fourth source contact **278**, the second pull-up device (PU-2) has a third source contact **276**, the third pass-gate device (PG-3) has a fifth source contact **280**, and the fourth pass-gate device (PG-4) has a sixth source contact **282**. As shown in FIG. 4A, by way of a contact via, the first source contact **272** is electrically coupled to the first bit line (BL-A) **230-1**. In a similar fashion, the second source contact **274** is electrically coupled to the second bit line (BL-B) **230-2**, the sixth source contact **282** is electrically coupled to the first bit line bar (BL-A bar) **232-1**, and the fifth source contact **280** is electrically coupled to the second bit line bar (BL-B bar) **232-2**. By means of contact vias, the third source contact **276** and the fourth source contact **278** are electrically coupled to a power line (Vdd) **238**. By way of a first Vss via **260**, the third long contact **244** is electrically coupled to a first complimentary power line (1.sup.st Vss) **234**. By way of a second Vss via **262**, the fourth long contact **246** is electrically coupled to a second complimentary power line (2.sup.nd Vss) **236**.

[0031] Referring still to FIG. 4A, each of the first WL-A landing pad **226-1**, the first bit line (BL-A) **230-1**, the first complimentary power line (1.sup.st Vss) **234**, the second bit line (BL-B) **230-2**, the first WL-B landing pad **228-1**, the power line (Vdd) **238**, the second WL-B landing pad **228-2**, the second bit line bar (BL-B bar) **232-2**, the second complimentary power line (2.sup.nd Vss) **236**, the first bit line bar (BL-A bar) **232-1**, and the second WL-A landing pad **226-2** extends lengthwise along the Y direction, parallel to the lengthwise directions of the active regions (**204-1~6**). In some embodiments, the first WL-A landing pad **226-1**, the first bit line (BL-A) **230-1**, the first complimentary power line (1.sup.st Vss) **234**, the second bit line (BL-B) **230-2**, the first WL-B landing pad **228-1**, the power line (Vdd) **238**, the second WL-B landing pad **228-2**, the second bit line bar (BL-B bar) **232-2**, the second complimentary power line (2.sup.nd Vss) **236**, the first bit line bar (BL-A bar) **232-1**, and the second WL-A landing pad **226-2** may be disposed in a first metal layer (M1).

[0032] Referring to FIG. 4B, when viewed along the Z direction, the DP SRAM cell **200** is rectangular in shape and the elongated along the X direction. In some embodiments, the DP SRAM cell **200** includes a length L along the X direction and a width W along the Y direction. A ratio of the length L to the width W may be between about 4 and about 7. The smaller width W may reduce the line length of bit lines and bit line bars to reduce ohmic drop (i.e., IR drop) along the length of the bit lines and bit line bars. By situating the first WL-B landing pad **228-1** between the n-type second active region **204-2** and the p-type third active region **204-3**, the spacing between the n-type second active region **204-2** and the p-type third active region **204-3** may be increased. Similarly, by situating the second WL-B landing pad **228-2** between the n-type fifth active region **204-5** and the p-type fourth active region **204-4**, the spacing between the n-type fifth active region **204-5** and the p-type fourth active region **204-4** may be increased as well. As a result, a first spacing S1 between

adjacent n-type active regions is smaller than a second spacing S2 between a n-type active region and an adjacent p-type active region. For example, the first active region **204-1** and the second active region **204-2**, both being n-type, are spaced part along the X direction by the first spacing S1. The fifth active region **204-5** and the sixth active region **204-6**, both being n-type, are spaced apart from one another along the X direction by the first spacing S1 as well. The n-type second active region **204-2** and the p-type third active region **204-3** are spaced apart from one another by the second spacing S2. The n-type fifth active region **204-5** and the p-type fourth active region **204-4** are spaced apart from one another by the second spacing S2. As described above with respect to FIG. 4A, each the first active region **204-1**, the second active region **204-2**, the fifth active region **204-5**, and the sixth active region **204-6** includes two fin structures or a two vertical stacks of channel members while each of the third active region **204-3** and the fourth active region **204-4** includes a single fin structure or a single vertical stack of channel members. Therefore, the first spacing S1 refers to the smallest fin-to-fin or stack-to-stack spacing between fins/stacks in the first active region **204-1** and counterparts in the second active region **204-2** or the smallest fin-to-fin or stack-to-stack spacing between fins/stacks in the fifth active region **204-5** and counterparts in the sixth active region **204-6**. Similarly, the second spacing S2 refers to the smallest fin-to-fin or stack-to-stack spacing between fins/stacks in the second active region **204-2** and the fin/stack in the third active region **204-3** or the smallest fin-to-fin or stack-to-stack spacing between fins/stacks in the fifth active region **204-5** and the fin/stack in the fourth active region **204-4**.

[0033] In some implementations, a ratio of the second spacing S2 to the first spacing S1 may be between about 1.05 and about 2.00. The greater second spacing S2 may provide benefits. For example, the greater second spacing S2 may allow larger spacing between the n-type well **1000** and the p-type well **2000** (including the first p-type well **2000-1** and the second p-type well **2000-2**), which may result in reduced well isolation leakage and increase device immunity to latch-up and soft-error-rate. For another example, the greater second spacing S2 may make room for a larger n-type well, which may lead to a more balance performance characteristics of the DP SRAM cell **200**. As can be seen from FIGS. 4A and 4B, in a DP SRAM cells **200**, the p-type well **2000** (including the first p-type well **2000-1** and the second p-type well **2000-2**) is greater than the n-type well **1000** in terms of area. Increasing the area of the n-type well helps balance the performance of the DP SRAM cell **200**. Moreover, because the first complimentary power line (1.sup.st Vss) **234** and the second complimentary power line (2.sup.nd Vss) **236** are the only metal line disposed between adjacent n-type active regions (between the first active region **204-1** and the second active region **204-2** or between the fifth active region **204-5** and the sixth active region **204-6**), the first complimentary power line (1.sup.st Vss) **234** and the second complimentary power line (2.sup.nd Vss) **236** may be wider to make room for larger first Vss via **260** and the second Vss via **262**, which may lead to improved performance of the DP SRAM cell **200**. Compared to the gate vias (**252**, **254**, **256**, and **258**), the first Vss via **260** and the second Vss via **262** have larger dimensions. For example, when both gate vias and Vss vias are substantially rectangular, gate vias have a first dimension D1 and the Vss vias have a second dimension D2 along the lone side. In some instances, a ratio of the second dimension D2 to the first dimension D1 may be between about 1.1 and about 4.

[0034] Various metal layers and contact vias of the DP SRAM cell **200** are illustrated in FIGS. 5 and 6. FIG. 5 is a schematic illustration of metal layers viewed along the Y direction and FIG. 6 is a schematic illustration of metal layers viewed along the X direction. For simplicity of illustrate, in FIGS. 5 and 6, the first, second, third, fourth, fifth and sixth active regions (**204-1~6** in FIG. 4A) may be representatively shown as an active region **204**; the gate structures of the 10 transistors (PU-1, PU-2, PG-1~4, and PD-1~4 in FIG. 4A) may be representatively shown as a gate structure G; the first, second, third, and fourth gate vias (**252**, **254**, **256**, and **258** in FIG. 4A) may be representatively shown as a gate via GV; source contacts (**272**, **274**, **276**, **278**, **280**, and **282** in FIG. 4A), long contacts (**240**, **242**, **244**, and **246** in FIG. 4A) may be representatively shown as C in FIG.

6; source/drain contact vias may be representatively shown as V_o; and the n-type well **1000** and p-type well **2000** (including the first p-type well **2000-1** and the second p-type well **2000-2**) may be representatively shown as SW. An isolation layer **205**, such as a shallow trench isolation (STI) layer, may be disposed between adjacent active regions **204** to provide isolation. Each of the active regions **204** may be disposed over an n-type well, such as the n-type well **1000** in FIG. 4A or over a p-type well, such as the p-type well **2000** (including the first p-type well **2000-1** and the second p-type well **2000-2**) in FIG. 4A. Metal lines in the first metal layer (M1) extend lengthwise along the Y direction. Metal lines in the second metal layer (M2) extend lengthwise along the X direction. Metal lines in the third metal layer (M3) extend lengthwise along the Y direction. Metal lines in the fourth metal layer (M4) extend lengthwise along the X direction. First vias (V1) electrically couple the first metal layer (M1) and the second metal layer (M2). Second vias (V2) electrically couple the second metal layer (M2) and the third metal layer (M3). Third vias (V3) electrically couple the third metal layer (M3) and the fourth metal layer (M4). As described above, the first WL-A landing pad **226-1**, the first bit line (BL-A) **230-1**, the first complimentary power line (1.sup.st V_{ss}) **234**, the second bit line (BL-B) **230-2**, the first WL-B landing pad **228-1**, the power line (V_{dd}) **238**, the second WL-B landing pad **228-2**, the second bit line bar (BL-B bar) **232-2**, the second complimentary power line (2.sup.nd V_{ss}) **236**, the first bit line bar (BL-A bar) **232-1**, and the second WL-A landing pad **226-2** may be disposed in a first metal layer (M1).

[0035] Because the metal lines in the first metal layer (M1), the second metal layer (M2), the third metal layer (M3), and the fourth metal layer (M4) either extends along the X direction or along the perpendicular Y direction, the metal lines in the DP SRAM cell **200** do not include complex shapes, resulting in an improved yield. As shown in FIGS. 3, 5 and 6, the X-direction-extending second metal layer (M2) and the fourth metal layer (M4) may be used as the first word-line (WL-A) and the second word-line (WL-B) to couple to the first word-line (WL-A) driver/selector **14** and the second word-line (WL-B) driver/selector **16**. The Y-direction extending first bit line (BL-A), first bit line bar (BL-A-bar), second bit-line (BL-B), and the second bit-line bar (BL-B-bar) may be coupled to the first Y-multiplexer (MUX) and sensing amplifier **18** or the second Y-multiplexer (MUX) and sensing amplifier **20**.

[0036] In some embodiments illustrated in FIG. 7, both the first word-line (WL-A) and the second word-line (WL-B) may be implemented in the second metal layer (M2). In these implementations, the first word-line (WL-A) may be coupled to the first WL-A landing pad **226-1** and the second WL-A landing pad **226-2** by way of two first vias V1; and the second word-line (WL-B) may be coupled to the first WL-B landing pad **228-1** and the second WL-B landing pad **228-2** by way of two first vias V1.

[0037] In some alternative embodiments shown in FIG. 8, the second word-line (WL-B) may be implemented in the second metal layer (M2) and the first word-line (WL-A) may be implemented in the fourth metal layer (M4). Because the first word-line (WL-A) and the second word-line (WL-B) are fabricated in different metal layers, a first metal jog feature **290** may be formed in the second metal layer (M2) to increase the width of the second word-line (WL-B) to reduce ohmic drop (i.e., IR drop). In some embodiments illustrated in FIG. 8, the second word-line (WL-B) has a first width W1 and the first metal jog feature **290** may boost the width to a second width W2. The second width W2 is greater than the first width W1. In these implementations, the second word-line (WL-B) may be coupled to the first WL-B landing pad **228-1** and the second WL-B landing pad **228-2** by way of two first vias V1. The first word-line (WL-A) may be coupled to the first WL-A landing pad **226-1** and the second WL-A landing pad **226-2** by way of two first vias V1, two contact pads **294** and **296** in the second metal layer (M2), two second vias V2, two contact pads **302** and **304** in the third metal layer (M3), and two third vias V3. The two contact pads **294** and **296** in the second metal layer (M2), two second vias V2, two contact pads **302** and **304** in the third metal layer (M3), and two third vias V3 help reroute the signal path between the fourth metal layer (M4) and the first WL-A landing pad **226-1** and the second WL-A landing pad **226-2**.

[0038] Further embodiments of metal layers may be illustrated in FIG. 9. Compared to the metal layer structures shown in FIG. 8, the metal layer structure in FIG. 9 further includes power mesh features may be incorporated in the metal layer structure shown in FIG. 8 to reduce ohmic drop (i.e., IR drop). As shown in FIG. 9, the first complimentary power line (1.sup.st Vss) 234 may be routed to the fourth metal layer (M4) by way of a first via V1, a contact pad 295 in the second metal layer (M2), a second via V2, a contact pad 306 in the third metal layer (M3), and a third via V3. Similarly, the second complimentary power line (2.sup.nd Vss) 236 may be routed to the fourth metal layer (M4) by way of a first via V1, a contact pad 297 in the second metal layer (M2), a second via V2, a contact pad 308 in the third metal layer (M3), and a third via V3. Because the contact pads 295 and 297 take up space in the second metal layer (M2), a second metal jog feature 291 in FIG. 9 in the second metal layer (M2) may be smaller than the first metal jog feature 290 in FIG. 8.

[0039] Although not intended to be limiting, one or more embodiments of the present disclosure provide benefits. For example, the present disclosure provides embodiments of a 10T SRAM cell that includes two pull-up devices (PU-1 and PU-2), fourth pull-down devices (PD-1, PD-2, PD-3, and PD-4), and four pass-gate devices (PG-1, PG-2, PG-3, and PG-4). The SRAM cell of the present disclosure includes word-line contact pads disposed between an n-type device and a p-type device to increase the spacing therebetween. The increased spacing allows reduction of latch up and reduction of soft-error-rate for improved performance. Additionally, the increased spacing between n-type and p-type devices allow room for a larger n-type well for a more balanced performance of the SRAM cell. To simplify routing structures and increase scalability, metal lines in one metal layer is orthogonal to those in an adjacent metal layer. An SRAM cell of the present disclosure includes a first long contact coupled to drains of the first pull-down device (PD-1), the second pull-down device (PD-2), and the first pull-up device (PU-1) and a second long contact are coupled to drains of the third pull-down device (PD-3), the fourth pull-down device (PD-4), and the second pull-up device (PU-2). The implementation of long contacts simplifies the routing structure.

[0040] Thus, in one aspect, the present disclosure provides an SRAM cell. The SRAM cell includes a first pull-down device (PD-1), a second pull-down device (PD-2), a first pass-gate device (PG-1), and a second pass-gate device (PG-2) disposed in a first p-well on a substrate, wherein active regions of the second pass-gate device (PG-2) and the second pull-down device (PD-2) are aligned along a first direction; a third pull-down device (PD-3), a fourth pull-down device (PD-4), a third pass-gate device (PG-3), and a fourth pass-gate device (PG-4) disposed in a second p-well on the substrate, wherein active regions of the third pass-gate device (PG-3) and the third pull-down device (PD-3) are aligned along the first direction; a first pull-up device (PU-1) and a second pull-up device (PU-2) disposed in an n-well disposed between the first p-well and the second p-well, where an active region of the first pull-up device (PU-1) extends along the first direction; and a first landing pad disposed between the second pull-down device (PD-2) and the first pull-up device (PU-1), wherein the first landing pad is electrically coupled to a gate structure of the second pass-gate device (PG-2) by way of a first gate via.

[0041] In some embodiments, the first gate via is disposed between the second pull-down device (PD-2) and the first pull-up device (PU-1). In some implementations, the SRAM cell may further include a second landing pad disposed between the third pull-down device (PD-3) and the second pull-up device (PU-2) and the second landing pad is electrically coupled to a gate structure of the second pass-gate device (PG-2) by way of a second gate via. In some embodiments, the second gate via is disposed between the third pull-down device (PD-3) and the second pull-up device (PU-2). In some embodiments, a drain of the first pull-down device (PD-1), a drain of the second pull-down device (PD-2), and a drain the first pull-up device (PU-1) are electrically coupled to a first long contact. In some instances, a drain of the third pull-down device (PD-3), a drain of the fourth pull-down device (PD-4), and a drain the second pull-up device (PU-2) are electrically coupled to a first long contact. In some implementations, a source of the first pull-down device (PD-1) and a

source of the second pull-down device (PD-2) are electrically coupled to a third long contact. In some instances, a source of the third pull-down device (PD-3) and a source of the fourth pull-down device (PD-4) are electrically coupled to a third long contact.

[0042] In another aspect, the present disclosure provides an SRAM cell. The SRAM cell may include a first pull-down device (PD-1), a second pull-down device (PD-2), a first pass-gate device (PG-1), and a second pass-gate device (PG-2) disposed in a first p-well on a substrate; a third pull-down device (PD-3), a fourth pull-down device (PD-4), a third pass-gate device (PG-3), and a fourth pass-gate device (PG-4) disposed in a second p-well on the substrate; and a first pull-up device (PU-1) and a second pull-up device (PU-2) disposed in an n-well disposed between the first p-well and the second p-well. The first pull-down device (PD-1), the second pull-down device (PD-2), and the first pull-up device (PU-1) share a gate structure extending along a first direction and a drain of the first pull-down device (PD-1), a drain of the second pull-down device (PD-2), and a drain the first pull-up device (PU-1) are electrically coupled to a first long contact.

[0043] In some embodiments, the SRAM cell may further include a first landing pad disposed between the second pull-down device (PD-2) and the first pull-up device (PU-1) and the first landing pad is electrically coupled to a gate structure of the second pass-gate device (PG-2) by way of a first gate via. In some implementations, the first gate via is disposed adjacent to the first long contact. In some instances, a drain of the third pull-down device (PD-3), a drain of the fourth pull-down device (PD-4), and a drain the second pull-up device (PU-2) are electrically coupled to a second long contact. In some embodiments, the SRAM cell may further include a second landing pad disposed between the third pull-down device (PD-3) and the second pull-up device (PU-2). The second landing pad is electrically coupled to a gate structure of the second pass-gate device (PG-2) by way of a second gate via. In some implementations, the second gate via is disposed adjacent to the second long contact.

[0044] In still another aspect, the present disclosure provides an SRAM cell. The SRAM cell includes a first pull-down device (PD-1), a second pull-down device (PD-2), a first pass-gate device (PG-1), and a second pass-gate device (PG-2) disposed in a first p-well on a substrate; a third pull-down device (PD-3), a fourth pull-down device (PD-4), a third pass-gate device (PG-3), and a fourth pass-gate device (PG-4) disposed in a second p-well on the substrate; and a first pull-up device (PU-1) and a second pull-up device (PU-2) disposed in an n-well disposed between the first p-well and the second p-well. The first pass-gate device (PG-1) and the first pull-down device (PD-1) share a first active region extending along a first direction. The second pass-gate device (PG-2) and the second pull-down device (PD-2) share a second active region extending along the first direction. A third active region of the first pull-up device (PU-1) extends along the first direction. The first active region and the second active region are spaced apart by a first spacing. The second active region and the third active region are spaced apart by a second spacing greater than the first spacing.

[0045] In some embodiments, a ratio of the second spacing to the first spacing is between 1.05 and 2. In some implementations, the first active region includes two fin structures, the second active region includes two fin structures, and the third active region includes a single fin structure. In some implementations, the SRAM cell may further include a first landing pad disposed between the second pull-down device (PD-2) and the first pull-up device (PU-1). The first landing pad is electrically coupled to a gate structure of the second pass-gate device (PG-2) by way of a first gate via and the first landing pad extends along the first direction. In some instances, the SRAM cell may further include a first long contact. A drain of the first pull-down device (PD-1), a drain of the second pull-down device (PD-2), and a drain the first pull-up device (PU-1) are electrically coupled to the first long contact. In some implementations, the first long contact extends along a second direction perpendicular to the first direction and the first landing pad spans over the first long contact.

[0046] The foregoing has outlined features of several embodiments. Those skilled in the art should

appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A static random access memory (SRAM) cell comprising: a first pull-down device and a first pass-gate device disposed in a first p-well on a substrate, wherein active regions of the first pass-gate device and the first pull-down device are aligned along a first direction; a second pull-down device disposed in a second p-well on the substrate, a first pull-up device and a second pull-up device disposed in an n-well disposed between the first p-well and the second p-well, where an active region of the first pull-up device extends along the first direction; and a first landing pad disposed between the first pull-down device and the first pull-up device, wherein the first landing pad is electrically coupled to a gate structure of the first pass-gate device by way of a first gate via.
